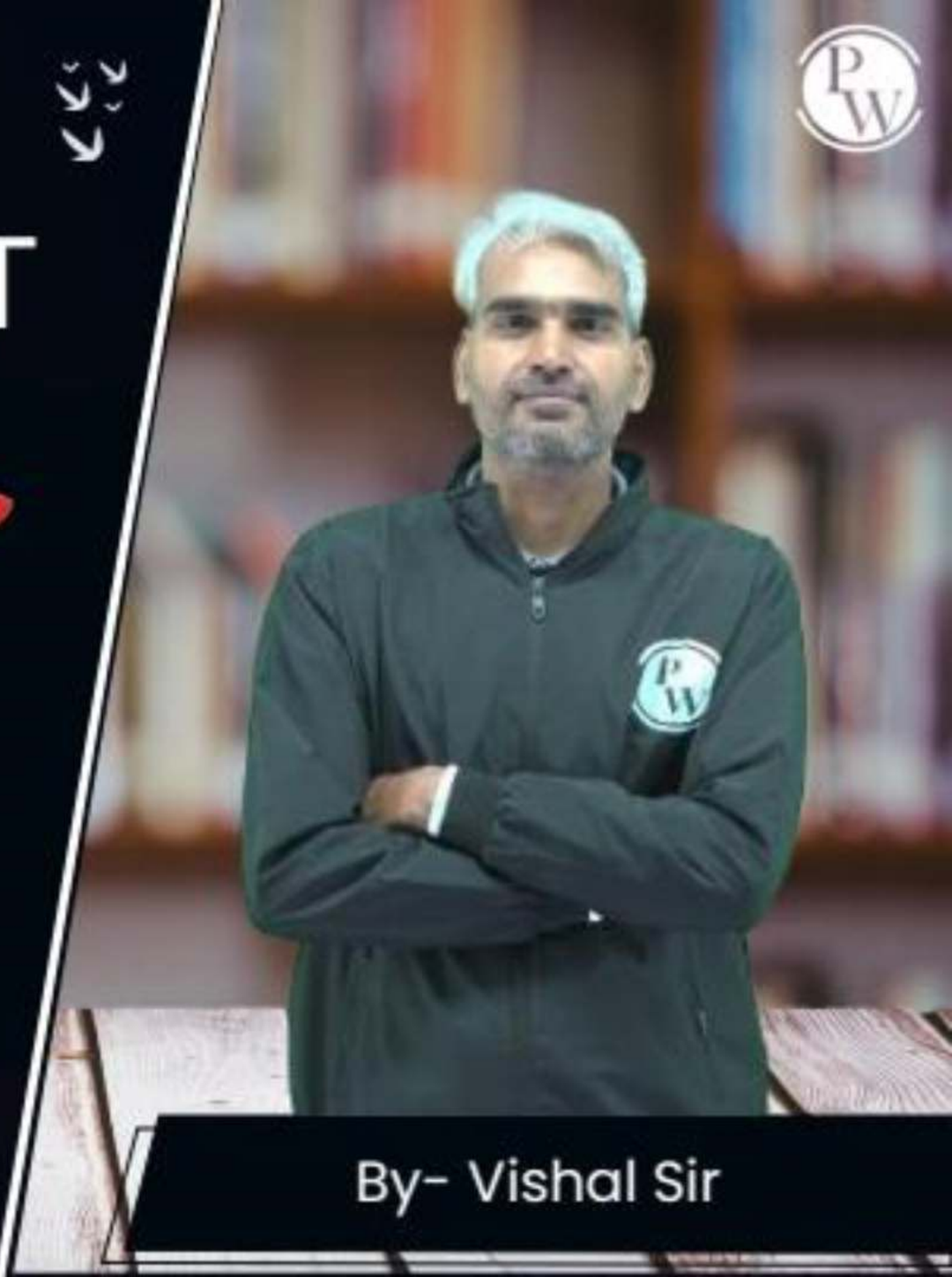
Computer Science & IT



Database Management System

Query Languages

Lecture No. 11



By- Vishal Sir

Recap of Previous Lecture



- **Topic** — Comparison with regular expression
- **Topic** — Introduction to relational calculus
- **Topic** — Tuple relational calculus

Slide

Topics to be Covered

Topic

Practice questions



Slide

#Q. Which of the following relational calculus expressions is not safe?

MCQ

A. $\{t \mid \exists u \in R1\ (t[A] = u[A]) \wedge \neg \exists s \in R2\ (t[A] = s[A])\}$

$\sim \exists S \in R_2\ (t[A] = S[A])$

$\forall S \in R_2\ (t[A] \neq S[A])$

B. $\{t \mid \forall u \in R1\ (u[A] = \text{“x”} \Rightarrow \exists s \in R2\ (t[A]=s[A] \wedge s[A] = u[A]))\}$

$S[A] = u[A] \wedge t[A] = S[A]$

$x : x$

C. $\{t \mid \neg(t \in R1)\}$

D. $\{t \mid \exists u \in R1\ (t[A] = u[A]) \wedge \exists s \in R2\ (t[A] = s[A])\}$

False

$t[A]=1$

$t[A]=2$

$t[A]=3$

$t[A]=4$

$u[A]$

R_1

A	B	C
1		
2		
3		
4		

R_2 S

C	A	D
	1	
	3	
	5	
	7	

#Q. Consider the relation employee (name, sex, supervisorName) with name as the key. Supervisor Name gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?

$\{e.name \mid employee(e)$
$\wedge (\forall x)\ [\neg employee(x) \vee x.supervisorName \neq e.name \vee x.sex = \text{“male”}]\}$

A. Names of employees with a male supervisor. Ans = Nutan

B. Names of employees with no immediate male subordinates Ans = {Ram, Mohan}

C. Names of employees with no immediate female subordinates Ans = {Mohan, Nutan, Geeta}

D. Names of employees with a female supervisor. Ans = {Ram, Mohan}

$$\{e.name \mid e \in employee \wedge \forall x \; [x \notin Employee \vee x.Supervisorname \neq e.name \vee x.Sex = male]\}$$

for tuples of Employee, this cond^{n} is permanently false

© e.name = Ram

Employee

Name	Sex	Supervisor.name
Ram	Male	Greeta
Mohan	Male	Nutan
Nutan	Female	Ram
Greeta	Female	NULL

e.name = Ram ✗

e.name = Mohan

E.name = Nutan

E.name = Greeta

x

Employee

Name	Sex	Supervisor.name	
Ram	Male	Greeta	✗
Mohan	Male	Nutan	✓
Nutan	Female	Ram	✓
Greeta	Female	NULL	✓

$\sim\sim[\forall x, [x \notin Employee\ e \vee x.Supervisorname \neq e.name \vee x.Sex = male]]$

$\{e.name \mid e \in employee \wedge \sim[\exists x [x \in Employee \wedge x.Supervisorname = e.name \wedge x.Sex = female]]\}$

Not — at least one tuple s.t.

#Q. Which of the following tuple relational calculus expression(s) is/are equivalent to $\forall t \in r(P(t))$?

I $\neg\exists t \in r(P(t))$

II $\exists t \notin r(P(t))$

III $\neg\exists t \in r(\neg P(t))$

IV $\neg\exists t \notin r(\neg P(t))$

$\forall t \in r(P(t))$ ⇒ for all tuples t belonging to R, $P(t)$ should be true

↳ i.e. There should not be at least one tuple in relation R, for which $P(t)$ is false

$$\sim\sim\{\forall t \in R\ (P(t))\}$$

$$\sim[\exists t \notin R\ (\sim P(t))\}$$

$$\sim(t \in R) \equiv t \notin R$$

$$\sim(\forall t \in R) = \exists t \in R$$

A. I only

B. II only

C. III only

D. III and IV only

#Q. Consider The Following Relational Scheme

- Student (school-id, sch-roll-no, sname, saddress)
- School (school-id, sch-name, sch-address, sch-phone)
- Enrolment (school-id, sch-roll-no, erollno, examname)
- ExamResult (Erollno, examname, marks)

Consider the following tuple relational calculus query

$$\{ t \mid \exists E \in \text{Enrolment}\ t = E.\text{school-id} \wedge$$

$$\frac{\{ x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t \wedge (\exists B \in \text{ExamResult}\ B.\text{erollno} = x.\text{erollno} \wedge B.\text{examname} = x.\text{examname} \wedge B.\text{marks} > 35 \}}{\{ x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t \}} * 100 > 35 \}$$

If a student needs to score more than 35 marks to pass an exam what does the query return?

- This ∧ return True
- school-ids from the Enrolment table will be present in the result provide
- No. of tuples from Specific School-id in which marks > 35
- R.H.S. of
- Total no. of Enrolments from School with Specific School-id

#Q. Consider The Following Relational Scheme

Student (school-id, sch-roll-no, sname, saddress)

School (school-id, sch-name, sch-address, sch-phone)

Enrolment (school-id, sch-roll-no, erollno, examname)

ExamResult (Erollno, examname, marks)

Consider the following tuple relational calculus query

$$\{ t \mid \exists E \in \text{Enrolment}\ t = E.\text{school-id} \wedge$$

$$|\{ x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t \wedge (\exists B \in \text{ExamResult}\ B.\text{erollno} = x.\text{erollno} \wedge$$

$$B.\text{examname} = x.\text{examname} \wedge B.\text{marks} > 35 \}| \div$$

$$|\{ x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t \}| * 100 > 35 \}$$

If a student needs to score more than 35 marks to pass an exam what does the query return?

$$\frac{m}{n} * 100 > 35$$

#Q.

A. The empty set

B. schools with more than 35% of its students enrolled in some exam or the other

C. schools with a pass percentage above 35% over all exams taken together

D. schools with a pass percentage above 35% over each exam

#Q. Consider The Following Relational Scheme

Student (school-id, sch-roll-no, sname, saddress)
School (school-id, sch-name, sch-address, sch-phone)
Enrolment (school-id, sch-roll-no, erollno, examname)
ExamResult (Erollno, examname, marks)

(A)for each school with more than 200 students appearing in exams, the name of the school and the number of 100s scored by its students

(B)for each school with more than 200 students in it, the name of the school and the number of 100s scored by its students

(C)for each school with more than 200 students in it, the name of the school and the number of its students scoring 100 in at least one exam

(D)nothing; the query has a syntax error

Question Continues in Next Slide

[MCQ: 2008: 2M]

#Q. What does the following SQL query output?

```
SELECT sch-name, COUNT (*)
FROM School C, Enrolment E, ExamResult R
WHERE   E.school-id = C.school-id
                AND
        E.examname = R.examname
                AND
        E.erollno = R.erollno
                AND
        R.marks = 100
                AND
        E.school-id IN
                (SELECT school-id
                FROM student GROUP BY school-id HAVING COUNT (*) > 200)

GROUP By school-id
```

Not in Group by ⇒ so invalid query

it will select School-id of the schools in which no. of students >200

H.W.

#Q. Consider a database that has the relation schemas
EMP(EmpId, EmpName, DeptId) and
DEPT(DeptName, DeptId).
Note that the DeptId can be permitted to a NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.

I. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \forall v \in DEPT\ (t[DeptId] \neq DeptId]))\}$

II. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \exists v \in DEPT\ (t[DeptId] \neq DeptId]))\}$

III. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \exists v \in DEPT\ (t[DeptId] = DeptId]))\}$

Which of the above queries are safe?

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

H.W.

#Q. [MCQ: 2009: 2M]

Consider the following relational schema:
Suppliers(sid:integer, sname:string, city:string, street:string)
Parts(pid:integer, pname:string, color:string)
Catalog(sid:integer, pid:integer, cost:real)

Consider the following relational query on the above database:

```
SELECT S.sname
        FROM Suppliers S
        WHERE S.sid NOT IN
                (SELECT C.sid
                        FROM Catalog C
                        WHERE C.pid NOT IN
                                (SELECT P.pid
                                        FROM Parts P
                                        WHERE P.color<> \'blue\'))
```

Question Continues in Next Slide

#Q. Assume that relations corresponding to the above schema are not empty. Which one of the following is the correct interpretation of the above query?

A. Find the names of all suppliers who have supplied a non-blue part.

B. Find the names of all suppliers who have not supplied a non-blue part.

C. Find the names of all suppliers who have supplied only blue parts.

D. Find the names of all suppliers who have not supplied only blue parts.



2 mins Summary

Topic Practice questions

Slide

THANK - YOU